**SEMESTER —VI (BATCH 2022, 2023 and onwards)**

| Sr.No | Course Category | Course Title | Course Code | Lecture(L)/ Tutorial(T)/ Practical(P) per week | | | Credits |
|---|---|---|---|---|---|---|---|
| | | | | L | T | P | |
| 1 | PC | Computer Networks | ECPC 308 | 3 | 1 | 0 | 3 |
| 2 | | Processor Design | ECPC 309 | 3 | 1 | 0 | 3 |
| 3 | | Wireless & Mobile Communication | ECPC 310 | 3 | 1 | 0 | 3 |
| 4 | PE | Program Elective –III | ECPE xxx | 3 | 1 | 0 | 3 |
| 5 | | Program Elective –IV | ECPE xxx | 3 | 1 | 0 | 3 |
| 6 | OE | Open Elective-II | ECOE xxx | 3 | 0 | 0 | 3 |
| 7 | PC | Processors Lab | ECPC 311 | 0 | 0 | 2 | 1 |
| 8 | | Embedded System Lab | ECPC 312 | 0 | 0 | 2 | 1 |
| 9 | | Machine Learning lab | ECPC 313 | 0 | 0 | 2 | 1 |
| 10 | NC | NCC/ Sports /Yoga | SWNC 101 | 0 | 0 | 4 | 2* |
| 11 | NC | NSS /Club/Technical Societies | SWNC102 | 0 | 0 | 4 | 2* |
| **Total** | | | | | | | **21** |

*Continuous Evaluation Model as per guidelines and the credit to be awarded at the end of 6th semester based on Cumulative performance up to 6th semester.

| **COMMUNICATION AND SIGNAL PROCESSING** | | | |
|---|---|---|---|
| **Sr.No** | **Course Category** | **Course Title** | **Course Code** |
| 1 | PE | Data Structure | ECPE-315 |
| 2 | | OFDM Systems | ECPE-316 |
| 3 | | Microstrip Antennas | ECPE-317 |
| 4 | | Adaptive Signal Processing | ECPE-318 |
| 5 | | Estimation & Detection | ECPE-319 |
| 6 | | Satellite Image Processing | ECPE-320 |
| 7 | | Computational Electromagnetics and its Applications | ECPE-321 |
| **DEVICES AND VLSI DESIGN** | | | |
| 1 | PE | Embedded System Fundamentals | ECPE-322 |
| 2 | | Analog IC Design | ECPE-323 |
| 3 | | Advanced Microcontroller Architecture | ECPE-324 |
| 4 | | Nano Electronics | ECPE-325 |
| 5 | | ASICs & FPGAs | ECPE-326 |
| **INTERNET OF THINGS** | | | |
| 1 | PE | Cryptography | ECPE-327 |
| 2 | | Data Science | ECPE-328 |
| 3 | | Image Processing | ECPE-329 |
| 4 | | Wireless Technology | ECPE-330 |
| 5 | | Advance Mobile Commn. | ECPE-331 |
| **Open Electives** | | | |
| 1 | OE | Wireless Communications | ECOE-303 |
| 2 | | Logic Design and Analysis using Verilog | ECOE-304 |

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

ECPCXX Computer Networks

Pre-requisite: Basics of communication systems

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 3 | 0 | 0 | 3 | 36 |

The objective of this course is to provide an understanding of theoretical aspects of computer networks, including the protocols involved in the exchange of information between communicating devices

**UNIT I**

Computer Network Topologies, Network Hardware, Network Software, OSI Model and TCP/IP protocol stack, ATM, Data communication fundamentals, wired physical layer, Wireless physical layer, and Physical layer based on telephone line. 9 hrs

**UNIT II**

Data Link layer design issues, Error detection & correction, Elementary Data Link protocols, Sliding Window Protocols, Example Data Link Protocols, Aloha Protocols, Wired MAC layer, IEEE 802.2: Logical Link control, Wireless MAC layer. 9hrs

**UNIT III**

Network layer services, Datagram and Virtual circuit services, Routing algorithms, Congestion control algorithms, Internetworking, Transport layer services, Elements of transport protocols, The Internet transport protocols: UDP & TCP. 9hrs

**UNIT IV**

Domain Name System, World Wide Web and HTTP, Electronic mail system, File Transfer protocol, Network security, Attacks on different network layers and their countermeasures. Use cases 9 hrs

**Reference Books:**


1. Tanenbaum A.S, "Computer Networks", Pearson, 4th ed., 2003.
2. Forouzan B.A, "Data Communications and Networking", Tata McGraw Hill, 4th ed. 2006.
3. Stallings W, "Data and Computer Communications", PHI, 9th ed., 2011.
4. Kurose & Ross K. W., "Computer Networking: A Top-Down Approach featuring the Internet", Pearson, 5th edition, 2010.


**Course outcomes:**

At the end of the course, the student will be able to:

CO1 Understand the computer network hardware and software.

CO2 Compare the OSI and TCP/IP protocol stacks.

CO3 Examine the protocols operating at different layers of network architecture.

CO4 Categorize the services offered by all layers of network's protocol stack.

CO5 Assess the cryptographic technique and Identify the sources of network security threats.

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT
ECPC38 PROCESSOR DESIGN

Pre-requisite:

| | |
|---|---|
| Course Code | ECPC38 |
| Course Title | POCESSOR DESIGN |
| Number of Credits | 03 |
| Prerequisites (Course code) | ECPC22, Computer Arch. |
| Course Type | PC |

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 3 | 0 | 0 | 3 | 42 |

Course Objectives

- To understand the architectures of popular processors and their area specific usage and advantages
- To learn the fundamentals of processor design
- To understand instruction level parallelism and multi-level pipelining
- To analyze Impact of workload, the degree of instruction-level parallelism, hardware parallelism , power consumption and technology on the efficiency of processors

UNIT I 8 hrs

CISC PROCESSOR DESIGN: Defining microprocessor, hardware flowchart, implementing from flowchart, exception, control store, microcode design.

UNIT II 10 hrs

RISC PROCESSOR DESIGN: Building datapath and controller, single cycle implementation, multi cycle implementation, pipelined implementation, exception and hazards handling. (Example: *DLX Processor*)

UNIT III 12 hrs

SUPERSCALAR PROCESSORS DESIGN: Superscalar organization, superscalar pipeline overview, VLSI implementation of dynamic pipelines, register renaming, reservation station, re-ordering buffers, branch predictor, and dynamic instruction scheduler etc.; simultaneous multi-threading (SMT) design. (Example: *Open SPARC T1*)

UNIT IV 12 hrs

MEMORY SYSTEM DESIGN. Application specific instruction set processor (ASIP) design. Dynamic reconfigurable processors (DRP). Impact of physical technology,

trends in power consumption, low power techniques, low voltage techniques, clock distribution.

Reference Books:


1. Nick Tredennick, *Microprocessor Logic Design*, Digital Press, 1987
2. D.A. Patterson and J.L. Hennessy, *Computer Organization and Design*, Morgan Kaufman Pub., N. Delhi, 2005
3. JP Shen and MH Lipasti, *Modern Processor Design*, MC Graw Hill, Crowfords ville, 2005
4. Mike Johnson, *Superscalar Microprocessor Design*, Prentice Hall, Englewood Cliffs, NJ, 1991
5. J.L. Hennessy, and D.A. Patterson, *Computer Architecture: A quantitative approach*, Fifth Edition, Morgan Kaufman Publication, 2012
6. A. Chandrakasan and W J Bowhill, and F. Fox, *Design of High-Performance Microprocessor Circuits*, IEEE Press, 2001


Course Outcomes

At the end of the course, the student will be able to:

CO1: Design basic processor and analyse the fundamentals of processor architecture
CO2: Choose application specific processor
CO3: Optimize parallelism and multi-level pipelining for performance and efficiency
CO4: Understand the design of instruction set , memory and reconfigurable processor

| | |
|---|---|
| Course Code | ECPC |
| Course Title | Wireless and Mobile Comm. |
| Number of Credits | 3 |
| Prerequisites (Course code) | |
| Course Type | PC |

## Subject : Wireless and Mobile Communication

### Course Learning Objectives

Course will provide an in depth understanding of the principles, performance and evolution of wireless communication standards (2G to 6G)

**Syllabus :**

### UNIT I

**Cellular Communications :** Introduction to Cellular Communications, Frequency reuse, Multiple Access, Technologies, Cellular Processes, GSM (SS7), Call Setup, Handover etc., Teletraffic Theory

**Wireless Communications and Diversity:** Fast Fading Wireless Channel Modeling, Rayleigh/Ricean Fading Channels, BER Performance in Fading Channels, Diversity modeling for Wireless Communications, BER Performance Improvement with diversity, Types of Diversity – Frequency, Time, Space.

### UNIT II

**Broadband Wireless Channel Modeling :** WSSUS Channel Modeling, RMS Delay Spread, Doppler Fading, Jakes Model, Autocorrelation, Jakes Spectrum, Impact of Doppler Fading

**CDMA :** Introduction to CDMA, Walsh codes, Variable tree OVSF, PN Sequences, Multipath diversity, RAKE Receiver, CDMA Receiver Synchronization.

### UNIT III

**OFDM:** Introduction to OFDM, Multicarrier Modulation and Cyclic Prefix, Channel model and SNR performance, OFDM Issues – PAPR, Frequency and Timing Offset Issues

**MIMO :** Introduction to MIMO, MIMO Channel Capacity, SVD and Eigen modes of the MIMO Channel, MIMO Spatial Multiplexing – BLAST, MIMO Diversity – Alamouti, OSTBC, MRT, MIMO-OFDM.

**UNIT IV**

**UWB (Ultrawide Band) :** UWB Definition and Features, UWB Wireless Channels, UWB Data Modulation, Uniform Pulse Train, Bit Error Rate Performance of UWB

**Evolution of Wireless Standards (2G-6G**): GPRS, WCDMA, LTE/ WiMAX, Cognitive Radios, IEEE wireless standards. Spectrum Sharing Framework for 6G-enabled IoT.

**Reference Books:**

1. Andrea Goldsmith, Wireless Communications: Cambridge University Press.

2. Theodore Rappaport, Wireless Communications: Principles and Practice, Prentice Hall.

3. Ezio Biglieri, MIMO Wireless Communications — Cambridge University Press.

4. Aditya K. Jagannatham, Principles of Modern Wireless Communication Systems: Mc GrawHill Education Course outcomes

At the end of the course student will be able to

1. Synthesis and analyze wireless and mobile cellular communication systems over different stochastic fading channels

2. Understand advanced multiple access techniques

3. Learn diversity reception techniques

4. Explore the need of MIMO/ OFDM as the pivoting technology for capacity maximization

5. Analyse the evolution of different wireless standards (2G TO 5G stds.) And the need

6. Contribute and meet the dynamic requirements of telecom companies

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

Processors Lab

PRACTICALS

SEMESTER-V, B.TECH

ECLR-32

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 0 | 0 | 2 | 1 | 14 |

List of Experiments

1. Arranging an Array of Bytes in Ascending /Descending Order.

2. Logic operations - Shift and rotate - Converting packed BCD to unpacked BCD, BCD to ASCII conversion.

3. String operations - Move block, Reverse string, String comparison, Length of string.

4. Find square root of a number & verify.

5. Generation of Sine, Triangular and Square wave using Interface with 8255 (PPI) and measurement of time period/frequency of sine, triangular and square wave.

6. Realize MOD-10 Up-Down Counter on 7-Segment display using Interface with 8255(PPI).

7. Interface Stepper Motor and controlling its RPM and direction of rotation.

8. ON/OFF control of SSR (Solid State Relay) using interface with 8255.

9. Interfacing of LM35/RTD temperature sensor with 8086 and display the temp value on LCD.

10. To interface traffic light system using 8086 & 8255.

# EMBEDDED SYSTEM LAB

## SUBJECT CODE: …….

**List of Experiments:**

**8051:**

Interfacing of high-power devices to Micro-controller port lines, LED, relays.

Program to control a stepper motor in direction, speed and number of steps using 8051.

Implementation of Digital FIR Filter on 8051 Microcontroller

DAC interfacing and generation of the ramp wave

ADC triggering through timer (On Chip Timer)

LCD interfacing and displaying a string

Keyboard interfacing takes input from the keypad and display it on LCD

Interface a LED matrix and display a number on the matrix.

Interfacing 4x4 switch matrix with the microcontroller

Serial Communication between microcontroller and PC

**ARM: LPC 2148 ARM 7 PROCESSOR**

ARM programming in C language using KEIL IDE

Implement the lighting and winking LEDs of the ARM I/O port via programming.

Changing ARM state mode by using MRS/MMSR instruction and specify a start address of the text segment by using command line.

To read analog voltage and display its digital equivalent on LCD using ARM.

**PIC: (PIC 16F877)**

Reading matrix keypad (DK) using PIC.

**AVR**

Stepper Motor Control Using ATMEGA-16 Microcontroller

Temperature control using ATmega16

**Course Outcomes:** At the end of this course, students should be able to:

CO1. Identify the functionality of development boards to implement embedded applications.

CO2. Compile bug free assembly or C language programs for microcontrollers to a required task.

CO3. Design an electronic circuit for diverse I/O devices used in real time embedded applications.

CO4. Develop a product with all sub systems of functional requirements in optimal hardware and software components.

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

**ECLR-33**
**Machine learning Lab**

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 0 | 0 | 2 | 1 | 2 |

**Brief Description:**
This lab is regarding the design and development of Machine learning algorithms for classification/ recognition of the data. It is also for developing and modelling the real time software for classification/ recognition of the data.

**Syllabus**

1. Development of k- nearest neighbors algorithm for classification of image data.

2. Implementation of k-means clustering algorithm for binary and multi-class classification of image data.

3. Development of expectation maximization (EM) algorithm for binary classification of the data and find the probabilities, means and variances of the respective classes.

4. Implement principle component analysis (PCA) technique on 2-D data and determine the Eigen vectors. Plot PCA space of the first two PCs.

5. Implement linear discriminant analysis (LDA) technique for data classification.

6. Apply PCA and LDA techniques for dimensionality reduction of feature vector.

7. Study of the SVM technique using MATLAB/C and apply this technique for data classification.

8. Study of the different deep learning techniques using MATLAB/C.

9. Design a feature map of a given data using convolution and pooling operation of convolutional neural network (CNN).

10. Minor project.

Text Books / Reference

1. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer

Course Outcomes: A student will develop different machine intelligence algorithms e.g. EM algorithm, SVM, Deep learning using MATLAB/C platform for real time application.

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

ECPEXX WIRELESS SENSOR NETWORKS

Pre-requisite: Computer networks

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 3 | - | - | 3 | 45 |

Course objective:

- To understand the fundamentals of wireless sensor networks and their application to critical real-time scenarios.
- To study the various protocols at different layers and their differences from traditional protocols.
- To understand the issues pertaining to sensor networks and the challenges involved in managing a sensor network.

Unit - I

Introduction to ad-hoc/sensor networks: Key definitions of ad-hoc/sensor networks, unique constraints, and challenges, advantages of ad-hoc/ sensor networks, driving applications, issues in ad-hoc wireless networks, issues in the design of sensor network, sensor network architecture, data dissemination, and gathering.

Unit – II

MAC Protocols: issues in designing MAC protocols for ad-hoc wireless networks, design goals, classification of MAC protocols, MAC protocols for sensor networks, S-MAC.
Routing Protocols: issues in designing a routing protocol, classification of routing protocols, table-driven, on-demand, hybrid, flooding, hierarchical, and power-aware routing protocols.

Unit - III

Topology control, clustering, Time-synchronization: clocks and the synchronization problem, time synchronization in WSNs, Time synchronization protocols, Localization: Ranging techniques, Ranging-based localization, Event-driven localization.

Unit – IV

Power management: Local power management aspects, Dynamic power management,
Security: challenges of security in WSNs, IEEE 802.15.4 Standard, and ZigBee.

Text Books / Reference:

1. C. Siva Ram Murthy, and B.S. Manoj, “Ad-hoc Wireless networks,” Pearson publication, 2008.
2. W. Dargie, and C. Poellabauer, “Fundamentals of wireless sensor networks: Theory and practice,” John Wiley and Sons Ltd., 2010.
3. I. F. Akylidiz, and M.C. Vuran, “Wireless sensor networks,” John Wiley and Sons Ltd., 2010.
4. Feng Zhao and Leonides Guibas, “Wireless sensor networks,” Elsevier publication, 2004.

Course Outcomes

CO 1: Technical knowledge in building a WSN network.

CO 2: Analysis of various critical parameters in deploying a WSN.

CO 3: Analysis of specific protocols related to medium access control, routing, and time-synchronization problems.

CO 4: Comprehend the study of localization strategies, power management, and security aspects.

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

ECPCXX IMAGE PROCESSING

Pre-requisite: Digital communication

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 3 | | | 3 | 42 |

**Brief Description about the course:**
Image processing is a rapidly growing field that deals with the use of digital images to various fields of life including medical, automation, etc. This course provides an in-depth introduction to the fundamental concepts, techniques, and tools used in image processing. The course covers the topics such as image acquisition, image processing, image compression, steganography, and digital watermarking. Students will learn how to apply these techniques to identify image tampering, detect forged images, and authenticate the origin of digital images.

**Course Learning Objectives**

The student should be able to understand and apply the various concepts of Digital Image Processing.

**Course Contents**

**UNIT I (10L)**

Introduction: The Origin of Digital Image Processing, Problems and Applications, Fundamental steps in image processing, Components of an image processing system. Transforms and their properties. Two Dimensional DFT, DCT. Hadamard, Slant, Haar and KL, Transforms and their properties
Image Perception: Image models, sampling & quantization, neighbors of a pixel, connectivity, labeling of connected components.

**UNIT- II (10L)**

Intensity Transformation and Spatial Filtering: Basic intensity transformation functions, Histogram processing, spatial filters, homomorphic filtering, generation of spatial marks. Fuzzy techniques for Intensity Transformation and Spatial Filtering.
Image Restoration and Reconstruction: "Noise models, Restoration in the presence of Noise only, Frequency domain filtering, Inverse filtering, Wiener filtering, Geometric mean filtering. Image reconstruction from projections.
Image Compression: Redundancy models, error free compression, Lossy compression, Some basic compression methods, Image compression standards.

**UNIT-III (6L)**

Color Image Processing: Color Models, Pseudocolor Image Processing, Bacics of full color image processing, color transformations, segmentation, smoothing and sharpening, Noise in color Images. VMF,VDF,etc.
Morphological Image Processing: Erosion, Dialation, Opening, Closing, Some basic morphological algorithms, Gray scale morphology.

**UNIT-IV (10L)+6L**

Image Segmentation: Detection of Discontinuity, Edge detection, Boundary detection, Thresholding, Regional oriented segmentation uses of motion in segmentation.
Representation and Description: Representation, Regional descriptors, Boundary descriptors, Image analysis, Pattern and their classes, Decision theoretical methods, Structural methods, Interpretation.
Copy paste attacks, its challenges and identification.
Noise removal from images depending on its nature.

**Reference Books:**


1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, Pearson, 3rd Edition, 2016.
2. Anil K. Jain, Fundamentals of Digital Image Processing, Pearson 2002.
3. Kenneth R. Castleman, Digital Image Processing, Pearson, 2006.
4. Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, 'Digital Image Processing using MATLAB', Pearson Education, Inc., 2000.
5. Hany Farid, 'Photo Forensics' The MIT Press, 2016.


**Course outcomes**

**At the end of the course student will be able to:**
**CO1: Understand and apply the Image filtering, Restoration, segmentation, etc.**
**CO2: Apply the vector approach to handle the color images.**
**CO3: Apply the 'Morphological operation' for various applications.**
**CO4: Develop the understanding about digital image forensics.**

GHANEKAR UMESH

ELECTRONICS & COMMUNICATION ENGINEERING DEPARTMENT

ECPC42 MICROWAVE THEORY AND DEVICES

Pre-requisite: ECPC21 (Electronic Devices and Circuits), ECPC24 (Fields and waves)

| L | T | P | Credits | Total contact hours |
|---|---|---|---|---|
| 3 | - | - | 3 | 42 |

Brief description about the course
The purpose of this course is to provide a comprehensive introduction to microwaves and principles of microwave devices.

Unit - I 10 hrs.
Introduction to Microwaves History and Applications, Review of Electromagnetics and Transmission Lines: Maxwell's equations, coaxial lines, rectangular and circular waveguides, microstrip lines.
Introduction to Microwave Measurements: Slotted Line, Tunable Detector, VSWR Meter, Power Meter, Spectrum Analyzer, Network Analyzer, Measurement of frequency, Power, Phase shift, VSWR and Impedance.

Unit – II 10hrs.
Microwave Components: S parameters, H, E and EH (Magic) plane Tee junction, Directional Couplers: Single Hole and Two Hole, Scattering Matrix of Directional Couplers, Waveguide Joints, Faraday Rotation: ferrite circulators, isolators and phase shifters, Microwave Attenuators.

Unit - III 10hrs.
Design considerations for microwave tubes, principle of operation of multi-cavity and reflex klystron, magnetron and traveling wave tube, Equivalent voltages and currents.

Unit – IV 12hrs.
Principles of Gunn diode, Tunnel Diode, IMPATT, TRAPPAT, BARRIT, PIN Diode and Schottky barrier diode, Microwave BJT, MESFET, HEMT and their applications.
Study of advanced microwave applications in industry and medical science: Microwave Imaging, Food Processing, Chemical Heating etc.

Text Books / Reference

1. ‘Microwave Engineering’ by David M. Pozar, John Wiley & Sons
2. ‘Microwave Devices and Circuits’ by Liao S.Y., Pearson Prentice Hall

3. ‘Foundations for Microwave Engineering’ by Robert E. Collin, Wiley-IEEE Press

Course Outcomes

CO1: Understand the basic concepts and applications of microwave systems.

CO2: Analyze, test and use various passive microwave components for different applications.

CO3: Understand the working of Klystron, TWT and Magnetron

CO4: Understand the principle of various microwave devices and their applications